Chapter 7 Review Questions

1. In infrastructure mode of operation, each wireless host is connected to the larger network via a base station (access point). If not operating in infrastructure mode, a network operates in ad-hoc mode. In ad-hoc mode, wireless hosts have no infrastructure with which to connect. In the absence of such infrastructure, the hosts themselves must provide for services such as routing, address assignment, DNS-like name translation, and more.

2. If the nodes are mobile, with connectivity changing among nodes, then they form a class of networks known as mobile ad hoc networks (MANETs). If the mobile nodes are vehicles, the network is a vehicular ad hoc network (VANET).

3. Path loss is due to the attenuation of the electromagnetic signal when it travels through matter. Multipath propagation results in blurring of the received signal at the receiver and occurs when portions of the electromagnetic wave reflect off objects and ground, taking paths of different lengths between a sender and receiver. Interference from other sources occurs when the other source is also transmitting in the same frequency range as the wireless network.

4. a) Increasing the transmission power
   b) Reducing the transmission rate

5. APs transmit beacon frames. An AP's beacon frames will be transmitted over one of the 11 channels. The beacon frames permit nearby wireless stations to discover and identify the AP.

6. The AP's SSID and MAC address.

7. APs transmit beacon frames. An AP's beacon frames will be transmitted over one of the 11 channels. The beacon frames permit nearby wireless stations to discover and identify the AP.

8. The process of scanning channels and listening for beacon frames by a wireless host is known as passive scanning. Broadcasting a probe frame that will be received by all APs within the wireless host's range is called active scanning.

9. It gives the sender explicit permission to send and also instructs the other stations not to send for the reserved duration.

10. No, there wouldn't be any advantage. Suppose there are two stations that want to transmit at the same time, and they both use RTS/CTS. If the RTS frame is as long as a DATA frames, the channel would be wasted for as long as it would have been wasted for two colliding DATA frames. Thus, the RTS/CTS exchange is only useful when the RTS/CTS frames are significantly smaller than the DATA frames.

11. Initially the switch has an entry in its forwarding table which associates the wireless station with the earlier AP. When the wireless station associates with the new AP, the new AP creates a frame with the wireless station's MAC address and broadcasts the frame. The frame is received by the switch. This forces the switch to update its forwarding table, so that frames destined to the wireless station are sent via the new AP.

12. While Bluetooth networks provide a "cable replacement" data rate of over a Megabit per second, Zigbee is targeted at lower powered, lower-data-rate, lower-duty-cycle applications than Bluetooth. Zigbee defines channel rates of 20, 40, 100, and 250 Kbps, depending on the channel frequency.

13. False

14. "Opportunistic Scheduling" refers to matching the physical layer protocol to channel conditions between the sender and the receiver, and choosing the receivers to which packets will be sent based on channel condition. This allows the base station to make best use of the wireless medium.

15. UMTS to GSM and CDMA-2000 to IS-95.

16. The data plane role of eNodeB is to forward datagram between UE (over the LTE radio access network) and the P-GW. Its control plane role is to handle registration and mobility signaling traffic on behalf of the UE.

    The mobility management entity (MME) performs connection and mobility management on behalf of the UEs resident in the cell it controls. It receives UE subscription information from the HHS.

    The Packet Data Network Gateway (P-GW) allocates IP addresses to the UEs and performs QoS enforcement. As a tunnel endpoint it also performs datagram encapsulation/decapsulation when forwarding a datagram to/from a UE.

    The Serving Gateway (S-GW) is the data-plane mobility anchor point as all UE traffic will pass through the S-GW. The S-GW also performs charging/billing functions and lawful traffic interception.

17. In 3G architecture, there are separate network components and paths for voice and data, i.e., voice goes through public telephone network, whereas data goes through public Internet. 4G architecture is a unified, all-IP network architecture, i.e., both voice and data are carried in IP datagrams to/from the wireless device to several gateways and then to the rest of the Internet.

    The 4G network architecture clearly separates data and control plane, which is different from the 3G architecture.

The 4G architecture has an enhanced radio access network (E-UTRAN) that is different from 3G's radio access network UTRAN.

18. No. A node can remain connected to the same access point throughout its connection to the Internet (hence, not be mobile). A mobile node is the one that changes its point of attachment into the network over time. Since the user is always accessing the Internet through the same access point, she is not mobile.

19. A permanent address for a mobile node is its IP address when it is at its home network. A care-of-address is the one its gets when it is visiting a foreign network. The COA is assigned by the foreign agent (which can be the edge router in the foreign network or the mobile node itself).

20. False

21. A GSM network's base station controller (BSC) will typically service several tens of Base Transceiver Stations (BTS). The role of the BSC is to allocate BTS radio channels to mobile subscribers, perform paging (finding the cell in which a mobile use is resident), and perform handoff of mobile users.

22. Anchor MSC is the MSC visited by the mobile when a call first begins; anchor MSC thus remains unchanged during the call. Throughout the call's duration and regardless of the number of inter-MSC transfers performed by the mobile, the call is routed from the home MSC to the anchor MSC, and then from the anchor MSC to the visited MSC where the mobile is currently located.

23. a) Local recovery
    b) TCP sender awareness of wireless links
    c) Split-connection approaches

Chapter 7 Problems

Problem 1

Output corresponding to bit d_1 = [–1, –1, 1, –1, –1, 1, 1, –1]
Output corresponding to bit d_0 = [1, 1, –1, 1, 1, –1, –1, 1]

Problem 2

Sender 2 output = [–1,1, –1, –1, –1,1, –1, –1]; [–1,1, –1, –1, –1,1, –1, –1]

Problem 3

The host will typically send a DHCP discovery message into the subnet via the AP in order to obtain an IP address on the subnet. Once the address is obtained, the rest of the world then views that host simply as another host with an IP address in that subnet.

Problem 4

No, because the code vectors are not orthogonal (dot product is not zero).

Problem 5

a) The two APs will typically have different SSIDs and MAC addresses. A wireless station arriving to the café will associate with one of the SSIDs (that is, one of the APs). After association, there is a virtual link between the new station and the AP. Label the APs AP1 and AP2. Suppose the new station associates with AP1. When the new station sends a frame, it will be addressed to AP1. Although AP2 will also receive the frame, it will not process the frame because the frame is not addressed to it. Thus, the two ISPs can work in parallel over the same channel. However, the two ISPs will be sharing the same wireless bandwidth. If wireless stations in different ISPs transmit at the same time, there will be a collision. For 802.11b, the maximum aggregate transmission rate for the two ISPs is 11 Mbps.

b) Now if two wireless stations in different ISPs (and hence different channels) transmit at the same time, there will not be a collision. Thus, the maximum aggregate transmission rate for the two ISPs is 22 Mbps for 802.11b.

Problem 6

Suppose that wireless station H1 has 1000 long frames to transmit. (H1 may be an AP that is forwarding an MP3 to some other wireless station.) Suppose initially H1 is the only station that wants to transmit, but that while half-way through transmitting its first frame, H2 wants to transmit a frame. For simplicity, also suppose every station can hear

every other station's signal (that is, no hidden terminals). Before transmitting, H2 will sense that the channel is busy, and therefore choose a random backoff value.

Now suppose that after sending its first frame, H1 returns to step 1; that is, it waits a short period of times (DIFS) and then starts to transmit the second frame. H1's second frame will then be transmitted while H2 is stuck in backoff, waiting for an idle channel. Thus, H1 should get to transmit all of its 1000 frames before H2 has a chance to access the channel. On the other hand, if H1 goes to step 2 after transmitting a frame, then it too chooses a random backoff value, thereby giving a fair chance to H2. Thus, fairness was the rationale behind this design choice.

Problem 7

A frame without data is 32 bytes long. The time to transmit a control frame (such as an RTS frame, a CTS frame, or an ACK frame) is (256 bits)/(12 Mbps) = 21 usec. The time required to transmit the data frame is (8256 bits)/(12 Mbps) = 688

DIFS + RTS + SIFS + CTS + SIFS + FRAME + SIFS + ACK
= DIFS + 3SIFS + (3 x 21 + 688) usec = DIFS + 3SIFS + 751 usec

Problem 8

a) 1 message/ 2 slots
b) 2 messages/slot
c) 1 message/slot

d) i) 1 message/slot
ii) 2 messages/slot
iii) 2 messages/slot

e) i) 1 message/4 slots
ii) slot 1: Message A→ B, message D→ C
slot 2: Ack B→ A
slot 3: Ack C→ D
= 2 messages/ 3 slots

iii)
slot 1: Message C→ D
slot 2: Ack D→C, message A→ B
slot 3: Ack B→ A
} Repeat

= 2 messages/3 slots

Problem 9

A node indicates to the access point that it will be going to sleep by setting the power-management bit in the header of an 802.11 frame to 1. A timer in the node is then set to wake up the node just before the AP is scheduled to send its beacon frame. Since the AP knows from the set power-transmission bit that the node is going to sleep, it (the AP) knows that it should not send any frames to that node, and will buffer any frames destined for the sleeping host for later transmission. A node will wake up just before the AP sends a beacon frame, and quickly enter the fully active state. The beacon frames sent out by the AP contain a list of nodes whose frames have been buffered at the AP. If there are no buffered frames for the node, it can go back to sleep. Otherwise, the node can explicitly request that the buffered frames be sent by sending a polling message to the AP. With an inter-beacon time of 100 msec, a wakeup time of 250 microseconds, and a similarly small time to receive a beacon frame and check to ensure that there are no buffered frames, a node that has no frames to send or receive can be asleep 99% of the time, resulting in a significant energy savings.

Problem 10

a) 10 Mbps if it only transmits to node A. This solution is not fair since only A is getting served. By "fair" it means that each of the four nodes should be allotted equal number of slots.

b) For the fairness requirement such that each node receives an equal amount of data during each downstream sub-frame, let n1, n2, n3, and n4 respectively represent the number of slots that A, B, C and D get.
Now,
data transmitted to A in 1 slot = 10t Mbits
(assuming the duration of each slot to be t)
Hence,
Total amount of data transmitted to A (in n1 slots) = 10t n1

Similarly total amounts of data transmitted to B, C, and D equal to 5t n2, 2.5t n3, and t n4 respectively.

Now, to fulfill the given fairness requirement, we have the following condition:

$10t\ n1 = 5t\ n2 = 2.5t\ n3 = t\ n4$

Hence,

$n2 = 2\ n1$
$n3 = 4\ n1$
$n4 = 10\ n1$

Now, the total number of slots is N. Hence,

$n1 + n2 + n3 + n4 = N$

i.e. n1+ 2 n1 + 4 n1 + 10 n1 = N
i.e. n1 = N/17

Hence,
n2 = 2N/17
n3 = 4N/17
n4 = 10N/17
The average transmission rate is given by:

(10t n1+5t n2+ 2.5t n3+t n4)/tN
= (10N/17 + 5 * 2N/17 + 2.5 * 4N/17 + 1 * 10N/17)/N
= 40/17 = 2.35 Mbps

c) Let node A receives twice as much data as nodes B, C, and D during the sub-frame.
Hence,
10tn1 = 2 * 5tn2 = 2 * 2.5tn3 = 2 * tn4
i.e. n2 = n1
n3 = 2n1
n4 = 5n1
Again,
n1 + n2 + n3 + n4 = N
i.e. n 1+ n1 + 2n1 + 5n1 = N
i.e. n1 = N/9

Now, average transmission rate is given by:
(10t n1+5t n2+ 2.5t n3+t n4)/tN
= 25/9 = 2.78 Mbps

Similarly, considering nodes B, C, or D receive twice as much data as any other nodes, different values for the average transmission rate can be calculated.

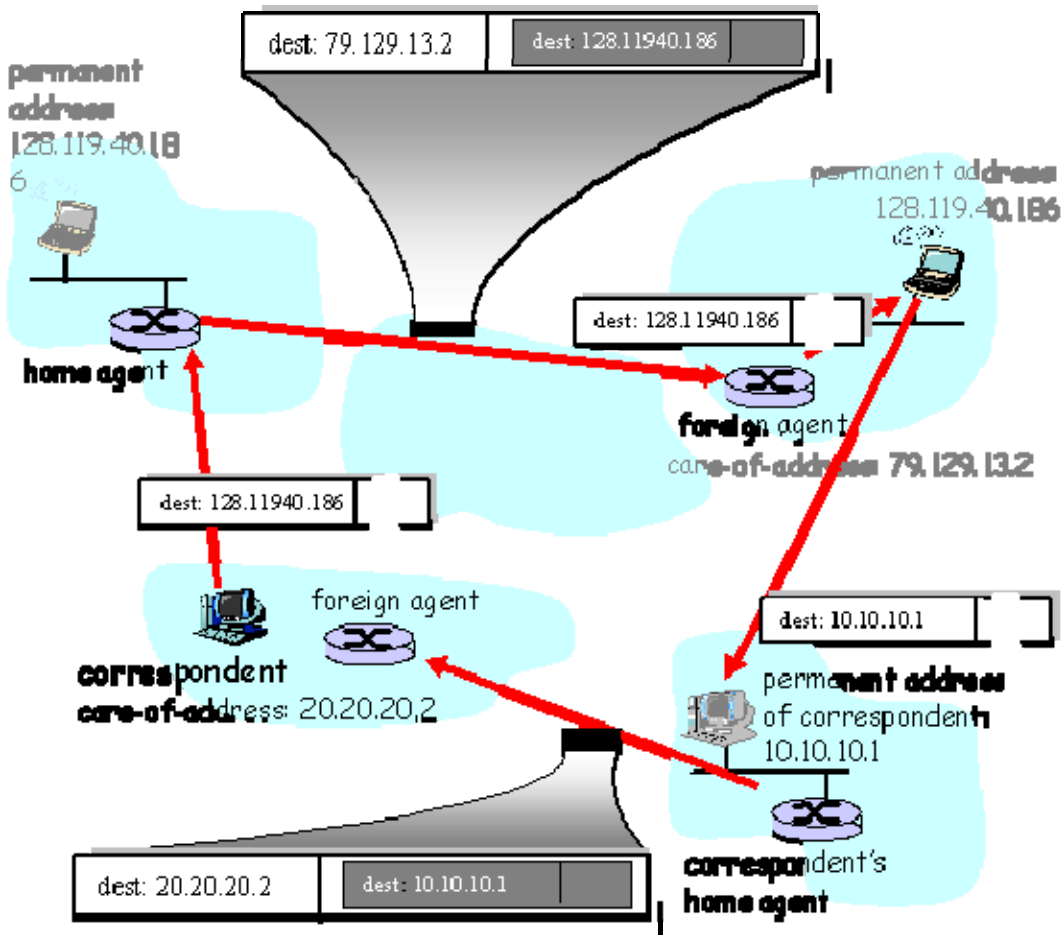
Problem 11

a) No. All the routers might not be able to route the datagram immediately. This is because the Distance Vector algorithm (as well as the inter-AS routing protocols like BGP) is decentralized and takes some time to terminate. So, during the time when the algorithm is still running as a result of advertisements from the new foreign network, some of the routers may not be able to route datagrams destined to the mobile node.

b) Yes. This might happen when one of the nodes has just left a foreign network and joined a new foreign network. In this situation, the routing entries from the old foreign network might not have been completely withdrawn when the entries from the new network are being propagated.

c) The time it takes for a router to learn a path to the mobile node depends on the number of hops between the router and the edge router of the foreign network for the node.

Problem 12



If the correspondent is mobile, then any datagrams destined to the correspondent would have to pass through the **correspondent's home agent**. The **foreign agent** in the network being visited would also need to be involved, since it is this foreign agent that notifies the correspondent's home agent of the location of the correspondent. Datagrams received by the correspondent's home agent would need to be encapsulated/tunneled between the correspondent's home agent and foreign agent, (as in the case of the encapsulated diagram at the top of Figure 6.23.

Problem 13

Because datagrams must be first forward to the home agent, and from there to the mobile, the delays will generally be longer than via direct routing. Note that it *is* possible, however, that the direct delay from the correspondent to the mobile (i.e., if the datagram is not routed through the home agent) could actually be smaller than the sum of the delay

from the correspondent to the home agent and from there to the mobile. It would depend on the delays on these various path segments. Note that indirect routing also adds a home agent processing (e.g., encapsulation) delay.

Problem 14

First, we note that chaining was discussed at the end of section 6.5. In the case of chaining using indirect routing through a home agent, the following events would happen:

- The mobile node arrives at A, A notifies the home agent that the mobile is now visiting A and that datagrams to the mobile should now be forwarded to the specified care-of-address (COA) in A.
- The mobile node moves to B. The foreign agent at B must notify the foreign agent at A that the mobile is no longer resident in A but in fact is resident in B and has the specified COA in B. From then on, the foreign agent in A will forward datagrams it receives that are addressed to the mobile's COA in A to the mobile's COA in B.
- The mobile node moves to C. The foreign agent at C must notify the foreign agent at B that the mobile is no longer resident in B but in fact is resident in C and has the specified COA in C. From then on, the foreign agent in B will forward datagrams it receives (from the foreign agent in A) that are addressed to the mobile's COA in B to the mobile's COA in C.

Note that when the mobile goes offline (i.e., has no address) or returns to its home network, the datagram-forwarding state maintained by the foreign agents in A, B and C must be removed. This teardown must also be done through signaling messages. Note that the home agent is not aware of the mobile's mobility beyond A, and that the correspondent is not at all aware of the mobile's mobility.

In the case that chaining is not used, the following events would happen:

- The mobile node arrives at A, A notifies the home agent that the mobile is now visiting A and that datagrams to the mobile should now be forwarded to the specified care-of-address (COA) in A.
- The mobile node moves to B. The foreign agent at B must notify the foreign agent at A and the home agent that the mobile is no longer resident in A but in fact is resident in B and has the specified COA in B. The foreign agent in A can remove its state about the mobile, since it is no longer in A. From then on, the home agent will forward datagrams it receives that are addressed to the mobile's COA in B.
- The mobile node moves to C. The foreign agent at C must notify the foreign agent at B and the home agent that the mobile is no longer resident in B but in fact is resident in C and has the specified COA in C. The foreign agent in B can remove its state about the mobile, since it is no longer in B. From then on, the home agent will forward datagrams it receives that are addressed to the mobile's COA in C.

When the mobile goes offline or returns to its home network, the datagram-forwarding state maintained by the foreign agent in C must be removed. This teardown must also be done through signaling messages. Note that the home agent is always aware of the mobile's current foreign network. However, the correspondent is still blissfully unaware of the mobile's mobility.

Problem 15

Two mobiles could certainly have the same care-of-address in the same visited network. Indeed, if the care-of-address is the address of the foreign agent, then this address would be the same. Once the foreign agent decapsulates the tunneled datagram and determines the address of the mobile, then separate addresses would need to be used to send the datagrams separately to their different destinations (mobiles) within the visited network.

Problem 16

If the MSRN is provided to the HLR, then the value of the MSRN must be updated in the HLR whenever the MSRN changes (e.g., when there is a handoff that requires the MSRN to change). The advantage of having the MSRN in the HLR is that the value can be provided quickly, without querying the VLR. By providing the address of the VLR Rather than the MSRN), there is no need to be refreshing the MSRN in the HLR.